FAULTY PANEL IDENTIFICATION

Summary Report

Distribution Automation Smart Grid – EC 9130

Group – 09

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OBJECTIVE OF WEEK 14

The objective of this week was to enhance the PV fault detection system to accurately detect:

- Irradiance-change faults (partial shading)
- Multiple faults occurring simultaneously

OBSERVED ISSUES

- Unable to detect irradiance variation faults in multiple modules
- Could only detect single fault at a time
- Incorrect module identification when multiple faults occur

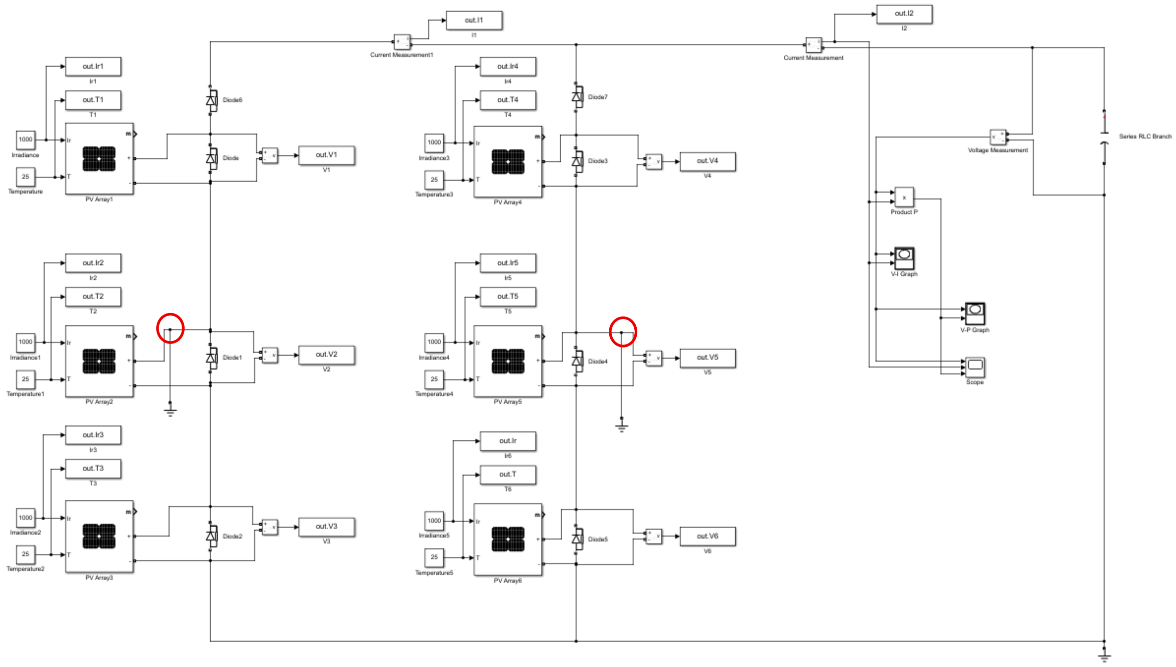


Fig 01: System when multiple Line-to-Ground fault condition is created.

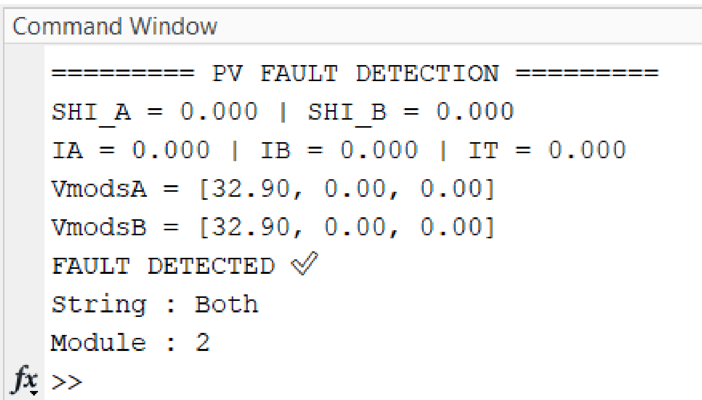


Fig 02: Old algorithm reported wrongly.

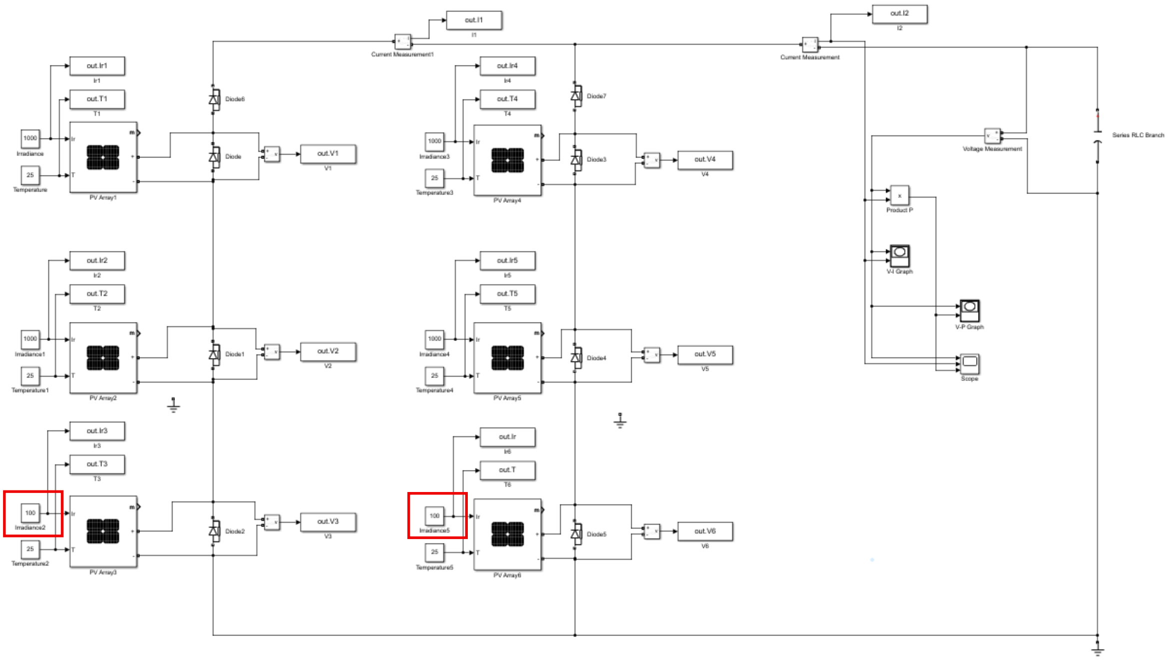


Fig 03: System when Irradiance change (Patial Shading) applied to multiple modules.

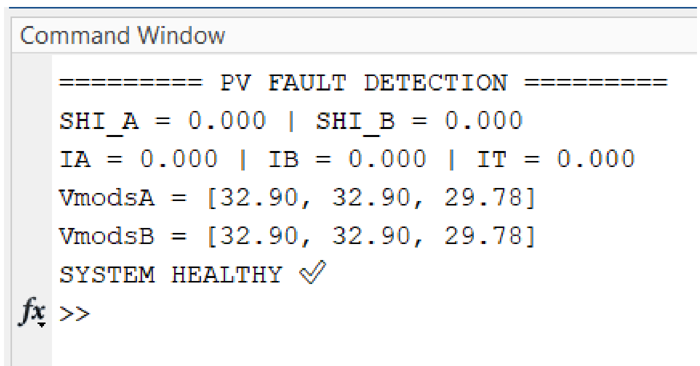


Fig 04: Old algorithm reported wrongly as System Healthy.

<u>ROOT CAUSE ANALYSIS</u>

1. Irradiance Fault Not Detected

   - Detection logic relied mainly on Voltage drops and Current variation
   - But Irradiance change does not always cause large voltage drop, Especially under low or zero current condition

2. Multiple Fault Issue
    - Previous logic used:

      min(VmodsA)

    - Only one module per string was identified
    - Other faulty modules were ignored

SOLUTION IMPLEMENTED

1. Irradiance-Based Fault Detection
    - Added module-wise irradiance comparison
    - Detection condition:

      G_module < (1 - threshold) * mean(G_string)

    - Combined with Voltage deviation and Electrical behavior
2. Multiple Fault Detection
    - Replaced single-index logic with:
        - loop-based detection
        - grouped fault identification
    - Implemented:

      candA = lowA | abnVA | abnGA;

    - Enabled:
        - detection of multiple faulty modules
        - correct identification per string
3. Grouped Fault Logic
    - Consecutive faulty modules handled as:
        - one fault region
        - first module reported as primary fault

RESULTS

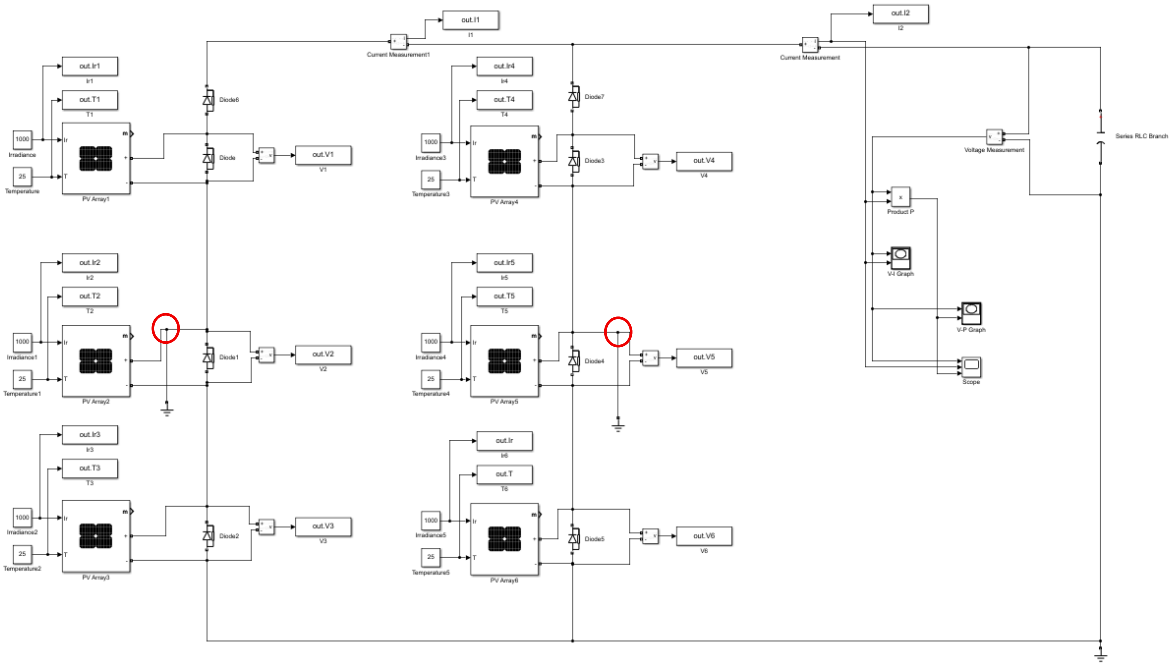


Fig 05: System when multiple Line-to-Ground fault condition is created

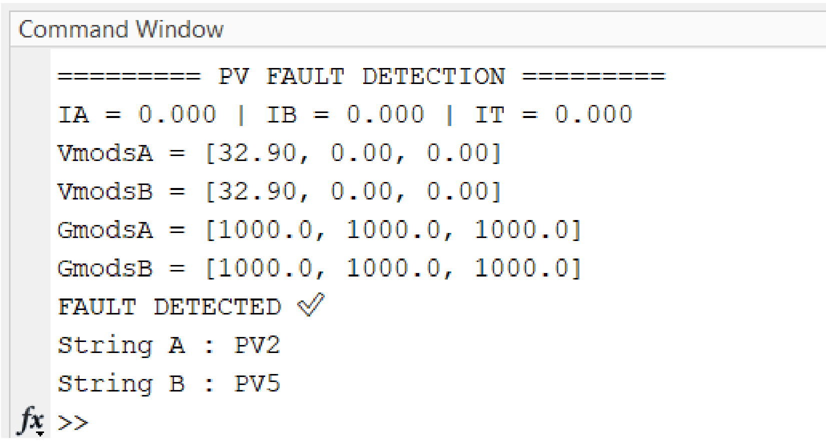


Fig 06: New algorithm reported correctly.

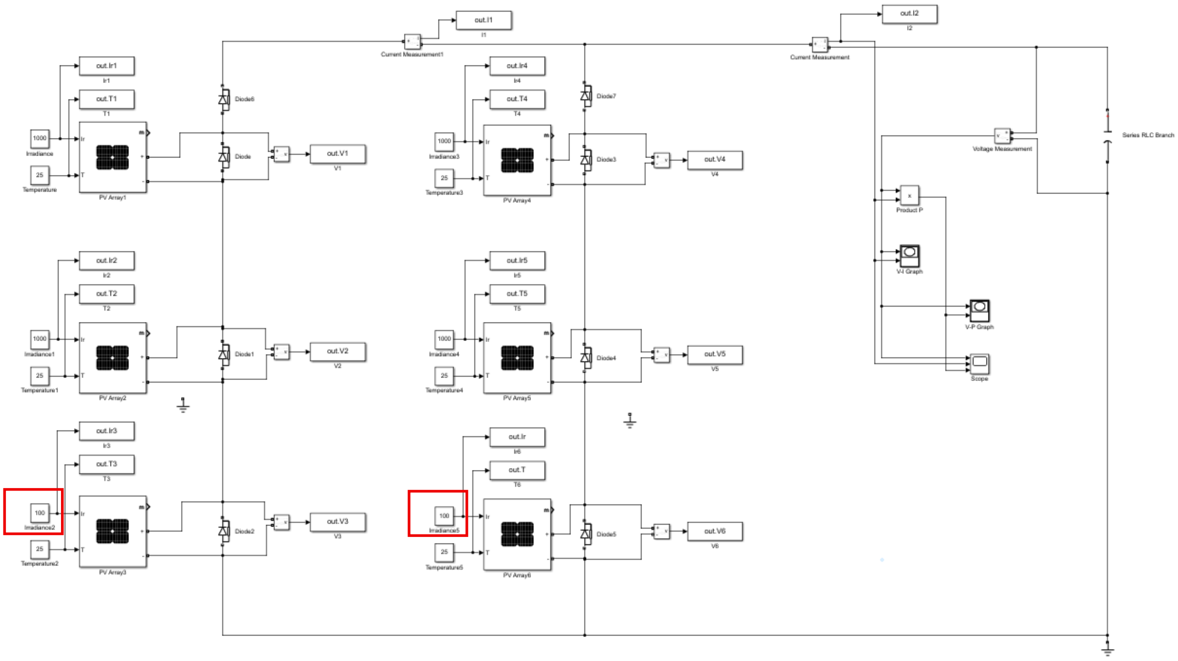
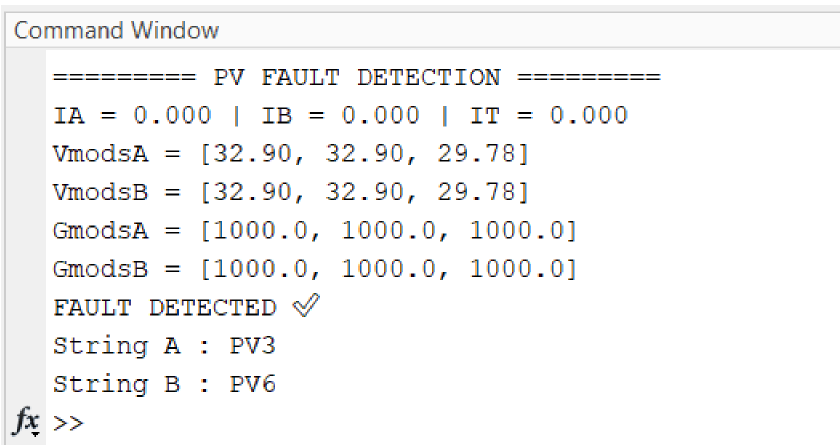


Fig07: System when Irradiance change (Patial Shading) applied to multiple modules.



Command Window

```
========= PV FAULT DETECTION =========
IA = 0.000 | IB = 0.000 | IT = 0.000
VmodsA = [32.90, 32.90, 29.78]
VmodsB = [32.90, 32.90, 29.78]
GmodsA = [1000.0, 1000.0, 1000.0]
GmodsB = [1000.0, 1000.0, 1000.0]
FAULT DETECTED ✓
String A : PV3
String B : PV6
fx >>
```

Fig 08: New algorithm reported correctly.

UPDATED DETECTION CAPABILITY

The system can now detect:

Fault Type	Detection Status
Healthy condition	Detected
Line to Ground Fault	Detected
Irradiance Change (Partial Shading)	Detected
Multiple Fault	Detected

SUMMARY OF WORK DONE

- Implemented irradiance-based detection logic
- Improved voltage deviation sensitivity
- Enabled multi-module fault identification

CONCLUSION

- The PV fault detection system was successfully enhanced to detect both electrical and environmental faults. The algorithm is now robust, scalable, and capable of handling real-world conditions such as partial shading and multiple simultaneous faults.